Column Generation and Constraint Generation

A linear program can have too many variables or constraints to include them all at once.

> **Keep a smaller LP. Use its solution to decide what to add.**

We will complete a cutting-stock example, then add violated constraints to a second LP. Pause at the practice prompts; answers follow on the next page.

- Column generation: cutting stock
- Constraint generation: a geometric example
- Proofs and optional decomposition connections

What is missing from the smaller LP?

Approach	Start with	Find and add
Column generation	A subset of variables	A variable that fails the reduced-cost test
Constraint generation	A subset of constraints	A constraint violated by the current solution

> Both methods repeatedly solve an LP. Their final stopping tests check the parts of the original model that were omitted.

A small model alone is not enough: we also need a reliable way to search the missing columns or rows.

Cut rolls into pieces while using fewer rolls

Piece type	Width	Demand
1	3	4 pieces
2	4	3 pieces
3	6	2 pieces

Each stock roll has width 10 and costs 1 unit. A cutting pattern specifies how many pieces of each type come from one roll.

A pattern $a = (a_1, a_2, a_3)$ is feasible when $3a_1 + 4a_2 + 6a_3 \leq 10$, with nonnegative integer piece counts.

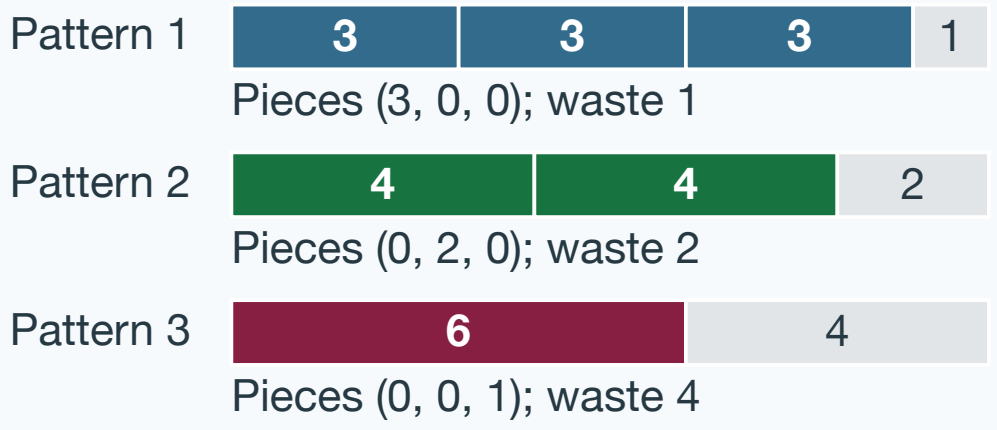
We solve the continuous LP: the amount assigned to each pattern may be fractional. An actual whole-roll cutting schedule is a separate problem.

Start with three simple cutting patterns

Let $x_j \geq 0$ be the amount of stock rolls assigned to pattern j.

Initial columns: $a^{(1)} = (3, 0, 0)$, $a^{(2)} = (0, 2, 0)$, $a^{(3)} = (0, 0, 1)$.

Each column lists the numbers of width-3, width-4, and width-6 pieces. These patterns provide a feasible starting plan.



Numbers inside pieces are widths. Gray is unused width.

Write the LP using only these three patterns

$$
\begin{aligned}
\min \quad & x_1 + x_2 + x_3 & \\
\text{s.t.} \quad & 3x_1 \geq 4 & \text{(width 3 demand)}, \\
& 2x_2 \geq 3 & \text{(width 4 demand)}, \\
& x_3 \geq 2 & \text{(width 6 demand)}, \\
& x_1, x_2, x_3 \geq 0. &
\end{aligned}
$$

> This is a **restricted master problem**: the original pattern LP, restricted to the columns currently included.

All demand constraints are present. Only the choice of patterns is restricted.

The first solution uses 29/6 rolls in the LP

$$x_1 = \frac{4}{3}, \qquad x_2 = \frac{3}{2}, \qquad x_3 = 2, \qquad z = \frac{4}{3} + \frac{3}{2} + 2 = \frac{29}{6}.$$

Produced pieces	Calculation	Demand
Width 3	$3(4/3) = 4$	4
Width 4	$2(3/2) = 3$	3
Width 6	$1(2) = 2$	2

The plan is feasible for the full pattern LP: set every omitted pattern quantity to zero. More available patterns may lower its cost.

The restricted dual tells us what to search for

Use a nonnegative dual variable p_i for demand row i. The restricted dual is

$$\begin{aligned} \max \quad & 4p_1 + 3p_2 + 2p_3 \\ \text{s.t.} \quad & 3p_1 \le 1, \quad 2p_2 \le 1, \quad p_3 \le 1, \qquad p \ge 0. \end{aligned}$$

$$p = \left(\frac{1}{3}, \frac{1}{2}, 1\right), \qquad 4p_1 + 3p_2 + 2p_3 = \frac{29}{6}.$$

> This certifies the restricted solution. A full dual certificate must also satisfy the inequality for every omitted pattern.

One omitted pattern breaks the current dual certificate

Pattern 4 cuts one width-4 piece and one width-6 piece: $a^{(4)} = (0, 1, 1)$.

$$p^\top a^{(4)} = \frac{1}{2} + 1 = \frac{3}{2} > 1.$$

Its reduced cost is $1 - p^\top a^{(4)} = -\frac{1}{2}$. The current dual vector violates the missing constraint $p_2 + p_3 \le 1$.

Add this column and solve the restricted LP again.

One roll: total width 10



Pieces (0, 1, 1); waste 0

Numbers inside pieces are widths. Gray is unused width.

Pricing: search for a column with negative reduced cost

Every roll costs 1. For any feasible pattern a, compute

$$\bar{c}(a) = 1 - p^\top a = 1 - (p_1 a_1 + p_2 a_2 + p_3 a_3).$$

> Find the largest $p^\top a$ over **all feasible patterns**. If it exceeds 1, that pattern has negative reduced cost.

This search is called **pricing**. Here the patterns are small enough to list; we do not need an integer-programming algorithm.

Negative reduced cost identifies a candidate. Degeneracy can still prevent an immediate strict improvement.

Add pattern 4 as one new variable

$$
\begin{aligned}
\min \quad & x_1 + x_2 + x_3 + x_4 \\
\text{s.t.} \quad & 3x_1 \geq 4, \\
& 2x_2 + x_4 \geq 3, \\
& x_3 + x_4 \geq 2, \qquad x \geq 0.
\end{aligned}
$$

> The new column is $a^{(4)} = (0, 1, 1)$: it adds no width-3 pieces, one width-4 piece, and one width-6 piece per roll.

Retain all earlier columns, including a column whose new quantity becomes zero.

Resolve: the value decreases from 29/6 to 23/6

$$x = \left(\frac{4}{3}, \frac{1}{2}, 0, 2\right), \qquad z = \frac{23}{6}.$$

Demand row	New production
Width 3	$3(4/3) = 4$
Width 4	$2(1/2) + 2 = 3$
Width 6	$0 + 2 = 2$

The new restricted dual solution is $p = (1/3, 1/2, 1/2)$. Its value is also $23/6$.

Use these updated dual variables in the next pricing search.

Your turn: price the next pattern

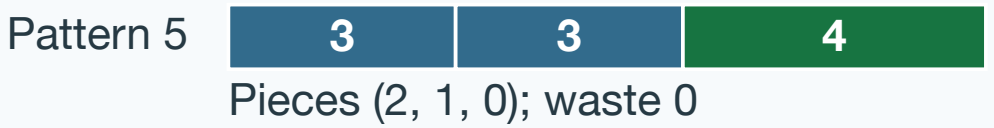
The current dual vector is $p = (1/3, 1/2, 1/2)$.

Consider pattern 5, $a^{(5)} = (2, 1, 0)$.

1. Check that the pieces fit within width 10.
2. Compute $p^\top a^{(5)}$.
3. Compute $1 - p^\top a^{(5)}$.
4. Should this column be added?

Pause for 5 minutes and write your four answers before checking the next page.

One roll: total width 10



Pieces (2, 1, 0); waste 0

Numbers inside pieces are widths. Gray is unused width.

Pattern 5 has reduced cost −1/6

$$3(2) + 4(1) + 6(0) = 10.$$

$$p^\top a^{(5)} = 2\left(\frac{1}{3}\right) + \frac{1}{2} = \frac{7}{6}, \qquad \bar{c}(a^{(5)}) = 1 - \frac{7}{6} = -\frac{1}{6}.$$

Add pattern 5. For these dual variables, it is the pattern with the largest value in the complete feasible-pattern list.

The full list and its coverage are shown after the next reoptimization.

The restricted LP now has five pattern variables

$$
\begin{aligned}
\min \quad & x_1 + x_2 + x_3 + x_4 + x_5 \\
\text{s.t.} \quad & 3x_1 + 2x_5 \geq 4, \\
& 2x_2 + x_4 + x_5 \geq 3, \\
& x_3 + x_4 \geq 2, \qquad x \geq 0.
\end{aligned}
$$

A new optimal solution is $x = (2/3, 0, 0, 2, 1)$, with $z = 11/3$.

Check production: $3(2/3) + 2 = 4$, $2 + 1 = 3$, and $2 = 2$.

Recover the new dual variables from the used patterns

The positive quantities are $x_1 = 2/3$, $x_4 = 2$, and $x_5 = 1$. Complementary slackness makes their dual inequalities active:

$$3p_1 = 1, \qquad p_2 + p_3 = 1, \qquad 2p_1 + p_2 = 1.$$

$$p_1 = \frac{1}{3}, \qquad p_2 = \frac{1}{3}, \qquad p_3 = \frac{2}{3}.$$

The restricted dual value is $4(1/3) + 3(1/3) + 2(2/3) = 11/3$. Now check *all* feasible patterns.

Check every pattern that could give a larger value

At $p = (1/3, 1/3, 2/3)$, the relevant pattern values are:

Pattern	Piece counts	Used width	$p^\top a$
1	(3,0,0)	9	1
2	(0,2,0)	8	$2/3$
3	(0,0,1)	6	$2/3$
4	(0,1,1)	10	1
5	(2,1,0)	10	1
6	(1,0,1)	9	1

> Every other feasible pattern can be filled out to one of these. Because the dual variables are nonnegative, adding a piece cannot decrease its dual value.

Thus no feasible pattern has value above 1. Pattern 6 has zero reduced cost and need not be added.

Matching bounds prove that we can stop

Our feasible plan uses $11/3$ rolls. Therefore the minimum over *all* patterns is at most $11/3$.

For every feasible pattern, $p^\top a \le 1$. Weak duality gives a lower bound $4p_1 + 3p_2 + 2p_3 = 11/3$.

$$\boxed{\text{Full LP optimum} = \frac{11}{3}.}$$

We did not need to include every pattern. We did need to check that no omitted pattern invalidated the dual certificate.

The complete column-generation loop

Solve the restricted LP

Read its dual variables

Search omitted columns

Add a violating column

Columns included	Restricted optimum	New column
1–3	$29/6$	4
1–4	$23/6$	5
1–5	$11/3$	None needed

> Stop only when the pricing search certifies that every omitted column satisfies its dual inequality.

In this minimization example, each restricted primal optimum is an upper bound on the full optimum.

The same idea for a general minimization LP

Let a_j be column j, c_j its cost, b the vector of constraint right-hand sides, and J the set of included column indices.

$$\min \sum_{j \in J} c_j x_j \quad \text{s.t.} \quad \sum_{j \in J} a_j x_j \geq b, \quad x_j \geq 0.$$

$$\max \; b^\top p \quad \text{s.t.} \quad a_j^\top p \leq c_j \; (j \in J), \quad p \geq 0.$$

Price omitted columns using $\bar{c}_j = c_j - a_j^\top p$. The stopping test is $\bar{c}_j \geq 0$ for every column.

The signs shown here belong to a min primal with ≥ rows and nonnegative variables.

Now start with fewer constraints

Consider the complete LP below. We will initially enforce only the variable bounds, then add the two other rows when needed.

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 \le 5, \quad 2x_1 + x_2 \le 6, \\ & 0 \le x_1 \le 4, \quad 0 \le x_2 \le 4. \end{aligned}$$

> Removing constraints enlarges the feasible set. For this maximization problem, the smaller model gives an **upper bound** on the true optimum.

All plotted regions refer to this same LP.

With only the box bounds, choose (4,4)

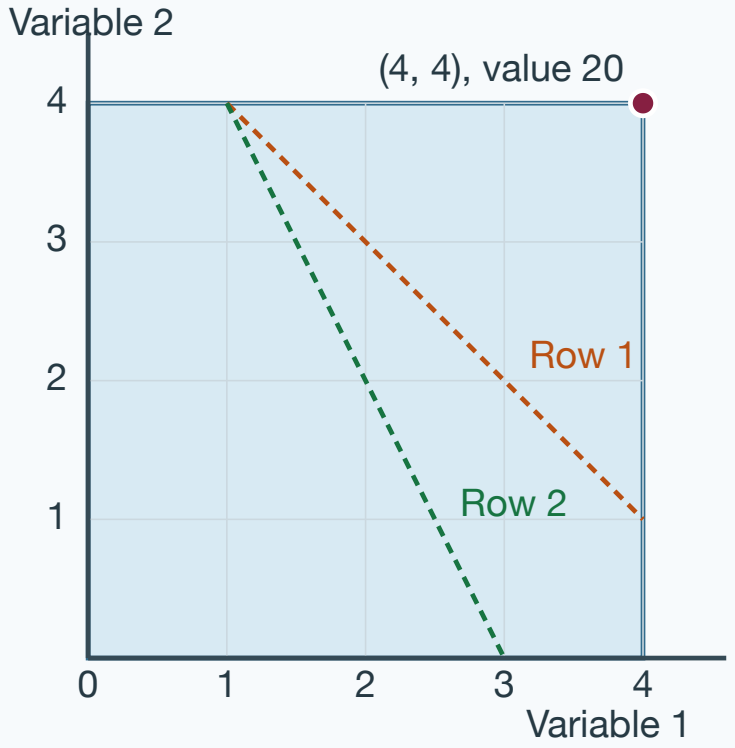
$$\max \; 3x_1 + 2x_2 \quad \text{s.t.} \quad 0 \le x_1, x_2 \le 4.$$

$$x = (4, 4), \qquad z = 20.$$

Check an omitted row:

$x_1 + x_2 = 8 > 5.$

The point is optimal for the box, but it is not feasible for the complete LP.



Horizontal: x_1; vertical: x_2. Orange: $x_1 + x_2 = 5$.
Green: $2x_1 + x_2 = 6$.

Add the violated row and solve again

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 \le 5, \qquad 0 \le x_1, x_2 \le 4. \end{aligned}$$

Variable 1 earns more per unit of the shared limit. Set it to 4, then use the remaining unit for variable 2.

$$x = (4, 1), \qquad z = 3(4) + 2(1) = 14.$$

The upper bound has decreased from 20 to 14. We must still test the omitted second row.

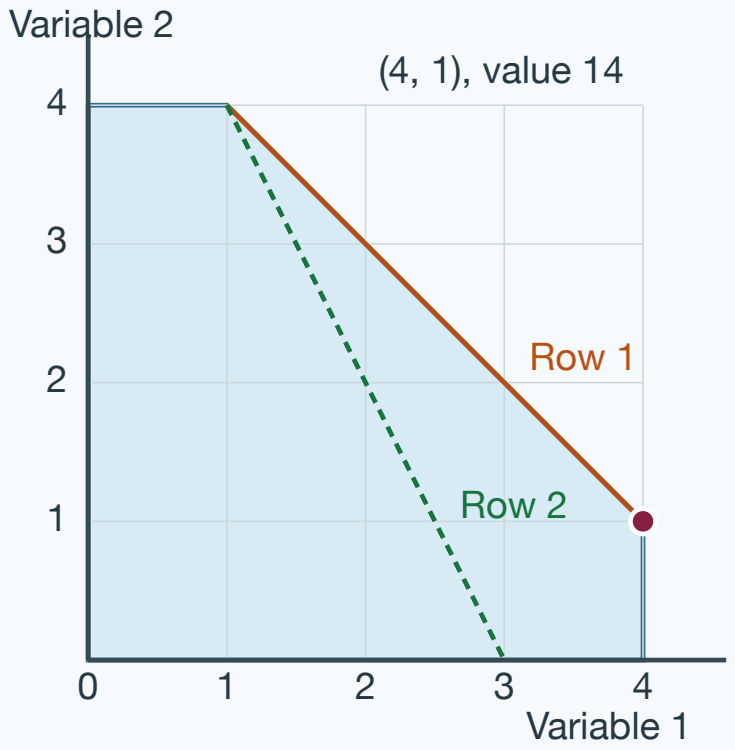
The smaller region still contains an infeasible optimum

Current solution: $x = (4, 1)$, value 14.

> Check the remaining row:
> $2x_1 + x_2 = 2(4) + 1 = 9 > 6.$

Add $2x_1 + x_2 \leq 6$. It removes the current point while preserving every feasible point of the full LP.

The orange line is enforced; the dashed green line is not yet enforced.



Horizontal: x_1; vertical: x_2. Orange: $x_1 + x_2 = 5$. Green: $2x_1 + x_2 = 6$.

After adding row 2, solve the complete LP

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 \le 5, \quad 2x_1 + x_2 \le 6, \\ & 0 \le x_1 \le 4, \quad 0 \le x_2 \le 4. \end{aligned}$$

The two added boundaries meet where $x_1 + x_2 = 5$ and $2x_1 + x_2 = 6$. Subtracting gives $x_1 = 1$, then $x_2 = 4$.

$$x = (1, 4), \qquad z = 3(1) + 2(4) = 11.$$

All original constraints are satisfied. The next page verifies optimality without relying only on the plot.

The current upper bound is now attained by a feasible point

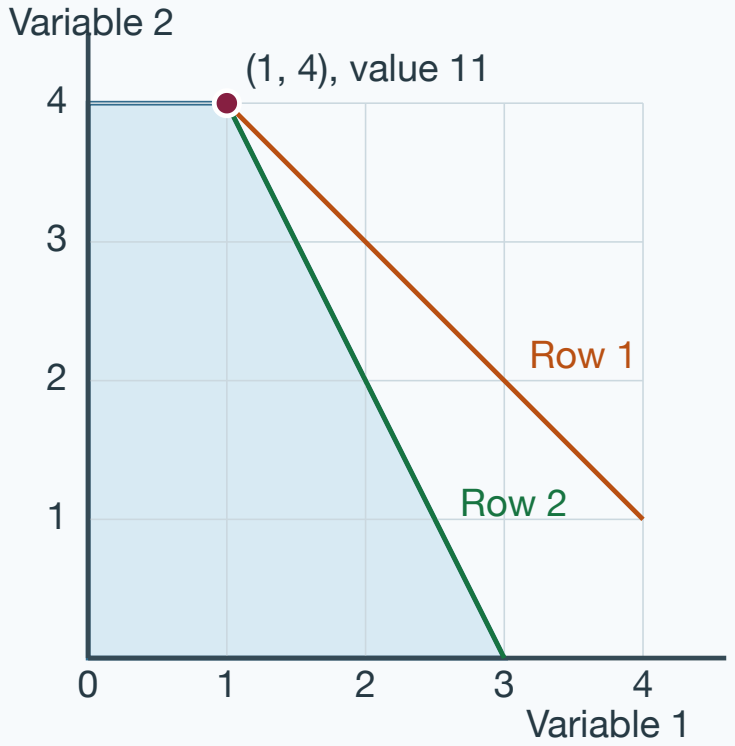
$$(x_1 + x_2) + (2x_1 + x_2) \leq 5 + 6.$$

$$3x_1 + 2x_2 \leq 11.$$

The point $(1, 4)$ is feasible and has value 11. It attains this upper bound.

The complete LP optimum is 11.

A stopping certificate checks feasibility for the full model, not just the constraints already drawn.

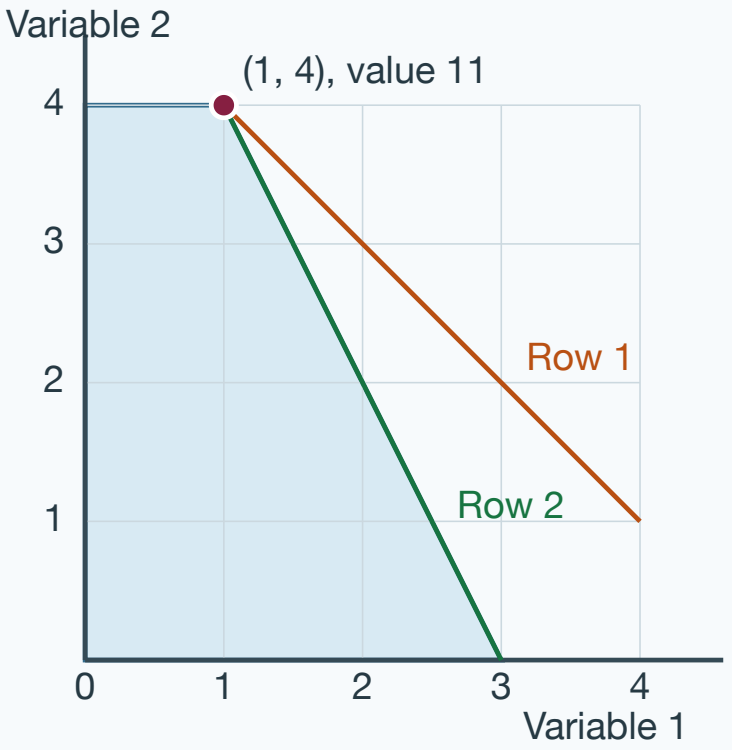


Horizontal: x_1; vertical: x_2. Orange: $x_1 + x_2 = 5$. Green: $2x_1 + x_2 = 6$.

Compare the three restricted problems

Rows added	Best point	Upper bound
None	(4,4)	20
Row 1	(4,1)	14
Rows 1 and 2	(1,4)	11

The upper bounds decrease. The final point satisfies every original row and attains the final bound.



Horizontal: x_1; vertical: x_2. Orange: $x_1 + x_2 = 5$. Green: $2x_1 + x_2 = 6$.

Separation: find a violated original constraint

Given the current solution x, search for a row $a_i^\top x \le b_i$ with $a_i^\top x > b_i$. This search is called **separation**.

If a violated row is found: add it and reoptimize.

If none exists: the current optimum is feasible for the full LP, so it is globally optimal.

This assumes the restricted LP has an optimal solution. A restricted LP can be unbounded even when the full LP is bounded; use a bounded starting relaxation or handle its improving ray.

For finitely many original rows, adding a new violated row each time cannot repeat forever.

Check the direction of the bound

A maximization LP has 500 constraints. We solve a restricted LP containing 20 of them. Its optimal value is 80.

1. Can the full LP have a feasible solution of value 85?
2. Suppose the current solution satisfies all 500 rows. What is the full optimum?
3. Suppose it violates one omitted row. Is 80 still an upper bound?

Pause for 4 minutes. Use the two feasible sets to explain your answers, then check the next page.

Fewer constraints give a larger feasible set

1. No. Every full-feasible solution was also allowed in the restricted problem, whose maximum was 80.

2. The full optimum is 80. The current point is full-feasible and attains the upper bound.

3. Yes. The upper bound remains valid. The point itself is infeasible, so add a violated row and reoptimize.

Columns and constraints: two different stopping checks

	Column generation here	Constraint generation here
Smaller problem	Min LP with fewer variables	Max LP with fewer constraints
Restricted solution	Feasible for the full primal	May violate omitted rows
What to check	All omitted dual inequalities	All omitted primal inequalities
What to add	A negative-reduced-cost column	A violated original row
At termination	Full dual certificate	Full primal feasibility

The bound direction follows from the objective sense and whether we restricted variables or removed constraints; do not memorize it without those details.

How does this relate to cutting-plane methods?

In **constraint generation**, we delay original constraints, then add violated ones. Every full-feasible solution satisfies each added row.

More generally, a **cutting-plane method** adds a valid inequality that removes the current point while preserving the feasible solutions of interest.

Our geometric example uses only original LP constraints. No integer-programming cuts are needed.

The important computational question is how to find a useful violated inequality efficiently.

What to take away

Column generation: solve a restricted LP, read its dual solution, price missing columns, add a column, and repeat until every reduced cost has the required sign.

Constraint generation: solve a relaxation, separate omitted rows, add a violated row, and repeat until the solution is feasible for the full LP.

Both methods need a complete stopping test. Searching a convenient subset is not enough unless it covers every possible violation.

Read BT §§6.1–6.3. Appendix: proofs and optional decomposition connections.

Appendix: proofs and optional connections

The first pages justify the stopping tests and complete the arithmetic for the cutting-stock example.

- Column-generation optimality proof
- Why the pattern list covers every case
- Verify both reoptimizations
- Constraint-generation optimality proof
- Optional: Dantzig–Wolfe and Lagrangian connections

Why the column-generation stopping test proves optimality

Consider $\min\{\sum_j c_j x_j : \sum_j a_j x_j \geq b,\ x \geq 0\}$. Let x^J, p^J be an optimal primal–dual pair for a feasible, bounded restricted master.

Extend x^J by zeros for omitted variables. It remains full-primal feasible. If pricing finds $a_j^\top p^J \leq c_j$ for every column, then p^J is full-dual feasible.

$$\underbrace{b^\top p^J}_{\text{full lower bound}} \leq z^* \leq \underbrace{\sum_{j \in J} c_j x_j^J}_{\text{full feasible value}} .$$

Restricted strong duality makes the two ends equal. Therefore both full solutions are optimal.

If the starting restricted master is infeasible, construct feasible starting columns or use a Phase I master; do not read dual prices from a nonexistent optimum.

Enumerate the maximal cutting patterns

A pattern can use at most one width-6 piece, since $2(6) > 10$.

Case	Patterns that cannot accept another piece
One width-6 piece	(1,0,1) or (0,1,1)
No width-6 piece; two width-4 pieces	(0,2,0)
No width-6 piece; one width-4 piece	(2,1,0)
No width-6 or width-4 pieces	(3,0,0)

> Every smaller feasible pattern is componentwise no larger than one of these five. With $p \geq 0$, its value cannot exceed that larger pattern's value.

The initial pattern (0,0,1) is feasible but not maximal. Keeping it as an initial column is harmless; pricing also checks the patterns that fill its remaining space.

Verify the first reoptimization with a matching bound

After adding pattern 4, use $x = (4/3, 1/2, 0, 2)$ and $p = (1/3, 1/2, 1/2)$.

$$3p_1 = 1, \quad 2p_2 = 1, \quad p_3 = \frac{1}{2} \le 1, \quad p_2 + p_3 = 1.$$

$$4p_1 + 3p_2 + 2p_3 = \frac{4}{3} + \frac{3}{2} + 1 = \frac{23}{6} = \sum_{j=1}^{4} x_j.$$

Primal feasibility was checked in the main sequence; these inequalities verify restricted dual feasibility. Equal objectives prove restricted optimality.

It is not yet a full certificate: pattern 5 has $2p_1 + p_2 = 7/6 > 1$.

Verify the second reoptimization and the whole-roll distinction

After adding pattern 5, $x = (2/3, 0, 0, 2, 1)$ produces exactly $(4, 3, 2)$, with objective $11/3$.

$$p = \left(\frac{1}{3}, \frac{1}{3}, \frac{2}{3}\right), \quad (p^\top a^{(1)}, \dots, p^\top a^{(6)}) = \left(1, \frac{2}{3}, \frac{2}{3}, 1, 1, 1\right).$$

This proves the continuous LP optimum is $11/3$. It does not permit us to physically cut a fraction of a roll.

For comparison, one copy of pattern 1, two of pattern 4, and one of pattern 5 use 4 whole rolls and produce $(5, 3, 2)$. Since any whole-roll plan needs at least $\lceil 11/3 \rceil = 4$ rolls, this plan is also best among whole-roll plans in this small example.

Why constraint generation proves optimality

Let P be the full feasible set and P_R the set defined by the retained rows. Then $P \subseteq P_R$. Assume $\hat{x}$ maximizes $c^\top x$ over P_R.

$$c^\top x \le c^\top \hat{x} \qquad \text{for every } x \in P.$$

If separation certifies $\hat{x} \in P$, then this upper bound is attained within P. Hence $\hat{x}$ is a full optimum.

Adding any violated original row preserves P and removes the current infeasible point. With finitely many rows, never discard added rows and never add a duplicate; only finitely many such additions are possible.

For a minimization problem with omitted constraints, the restricted optimum is instead a lower bound.

Optional: replace points by combinations of columns

Suppose a bounded polyhedron Q has vertices $v^1, \dots, v^K$. Every $x \in Q$ can be written as

$$x = \sum_{k=1}^{K} \lambda_k v^k, \qquad \sum_{k=1}^{K} \lambda_k = 1, \qquad \lambda_k \geq 0.$$

> Example: $Q = \{x \geq 0 : x_1 + x_2 \leq 2\}$ has vertices $(0,0), (2,0), (0,2)$. The point $(0,1)$ is $\frac{1}{2}(0,0) + \frac{1}{2}(0,2)$.

Dantzig–Wolfe decomposition uses such combinations to keep the constraints defining Q together and put the remaining constraints in a master LP. Those remaining constraints are called linking constraints.

Optional: the master LP and its pricing problem

Start with $\min\{c^\top x : Dx \geq d,\ x \in Q\}$, where Q is a nonempty bounded polyhedron. Substitution gives

$$\min_{\lambda \geq 0} \sum_k (c^\top v^k)\lambda_k \quad \text{s.t.} \quad \sum_k (Dv^k)\lambda_k \geq d, \quad \sum_k \lambda_k = 1.$$

Use $p \geq 0$ for the linking rows and unrestricted α for the sum-to-one row. Column k has reduced cost

$$c^\top v^k - p^\top D v^k - \alpha.$$

> Find the smallest reduced cost by solving $\min_{x \in Q} (c - D^\top p)^\top x - \alpha$. A linear objective on bounded Q attains its minimum at a vertex.

Optional: the same subproblem gives a Lagrangian bound

For $\min\{c^\top x : Dx \geq d,\ x \in Q\}$, choose $p \geq 0$ and define

$$L(x, p) = c^\top x + p^\top (d - Dx).$$

For every original feasible x, $d - Dx \leq 0$, so $L(x, p) \leq c^\top x$. Consequently,

$$p^\top d + \min_{x \in Q} (c - D^\top p)^\top x \ \leq\ \text{original minimum}.$$

This lower-bound calculation uses the same optimization over Q as column pricing. It explains the connection between continuous-LP decomposition and relaxing linking constraints.